

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8512**

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block: 401	Seat No: 2022600035
Course Section: 1		

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Name of the candidate. Nair Anurag Shivdas

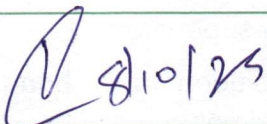
Candidate's UID number:

2	0	2	2	6	0	0	0	3	5
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Examination class room number:

4	0	1
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Invigilator's signature with date:



(To be filled in by the candidate)

EXAMINATION: BE BTECH
(For example FY MCA/FY BTECH)

SEMESTER : 7

PROGRAM: AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 8/10/25 Wed

COURSE NAME: DL

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



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	Q.2b	① Undercomplete autoencoders are the encoders with the dimension of bottleneck less than the input dimension. ② It learns to generalize the features of the input data into a compressed latent space. ③ It is used when data needs to be compressed and then reconstructed to the original form. If x is the original image & $\hat{x}$ the reconstructed image MSE would be $MSE = \ x - \hat{x}\ ^2$ ④ Encoder does processing of I/P $z = f(x)$$v = g(z)$$y = g(f(x))$ ⑤ Since the input of the decoder has lesser dimension as compared to the original, it works equivalent to PCA. ⑥ In a overcomplete autoencoder without regularization, the size of bottleneck is equal or larger than the input size. ⑦ Thus it tend to just duplicate the input rather than learning to



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generalize the features.
 (e) Output $\hat{x}$ would be given by

$$\hat{x} = \begin{bmatrix} I_n \\ 0 \end{bmatrix} [x \ 0] \begin{bmatrix} I_n \\ 0 \end{bmatrix}$$

where x is input &
 I_n is identity matrix

$$\hat{x} = x \cdot I_n + 0 \cdot 0$$

$$\hat{x} = x$$

$$MSE = |x - \hat{x}|^2 = 0$$

Therefore without regularization
 it tends to learn the identity
 function.


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- Q-3b
- ① GRU is updation of LSTM which has a streamline state instead of ~~se~~ separate hidden state and cell state.
 - ② Two gates work in a GRU namely update gate and reset gate.
 - ③ The ~~forget~~ update gate decides what part of old information and what part of new information to add.
 - ④ The reset gate decides what part of information is to be removed.

The ~~forget~~^{reset} gate work on sigmoid of previous hidden & current input

$$r_t = \sigma [W_r [h_{t-1}, x_t] + b_r]$$

$h_{t-1} \rightarrow$ previous hidden state

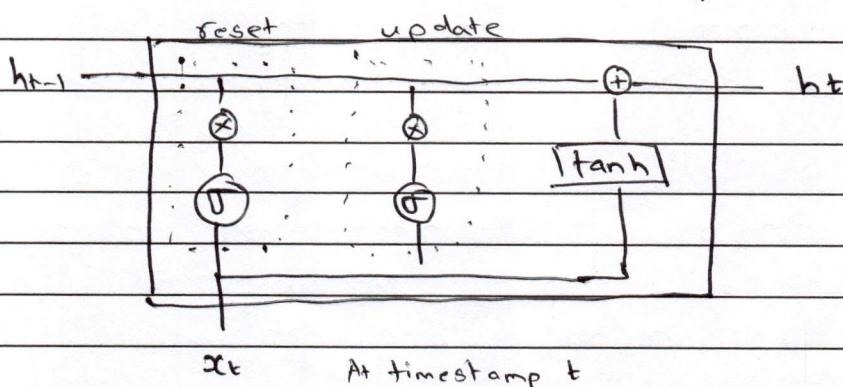
$x_t \rightarrow$ current input

$b_r \rightarrow$ bias at ~~forget~~ reset gate.

Update gate is given

$$z_t = \sigma [W_z [h_{t-1}, x_t] + b_z]$$

$b_z \rightarrow$ bias at update gate



The update gate is combination of previous hidden state and candidate hidden state.

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⑤ GRUs have lesser parameters as compared to those of LSTMs.

⑥ LSTMs have 3 gates i/p, o/p & forget so they have 4 parameters to work on (1 bias) & whereas for GRUs it is 3.
Now for same i/p size

$$\text{LSTM} = 4 \times n \times m \times b$$

$$\text{GRU} = 3 \times n \times m \times b$$

⑦ Thus GRUs would be more efficient and give faster results which can be used in real world applications where faster result are lesser control over memory is required.

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Q.1a

 η = learning rate

$$F(w) = \frac{1}{2} |xw - y|^2$$

$$z = -\eta \frac{\partial E}{\partial w}$$

$$\frac{\partial E}{\partial w} = \frac{\partial}{\partial w} \left[\frac{1}{2} |xw - y|^2 \right]$$

$$E = \frac{1}{2} \sum |t_d - o_d|^2 = \frac{1}{2} |xw - y|^2$$

$$\frac{\partial E}{\partial w} = \frac{1}{2} \frac{\partial}{\partial w} |xw - y|^2$$

$$= \frac{1}{2} \cdot 2 (xw - y) \frac{\partial (xw - y)}{\partial w}$$

$\therefore$ o/p var. is always constant

$$y = w \cdot 0 + wsc_1 + wsc_2$$

for y at a single data point (t)

$$\frac{\partial E}{\partial w} = (xw - y) (-x_t)$$

$$\therefore z = \eta (xw - y) \cdot x_t$$



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Q.2a

$$\text{I/P size} = H \times W \times C$$
$$\text{No. of Filter} = K$$
$$\text{Filter size} = F \times F$$

Assuming padding ~~and~~ as 0 & stride as 1
for a standard 2D convolution

$$\text{No. of multiplications} = F \times F \times (H-1) \times (W-1)$$

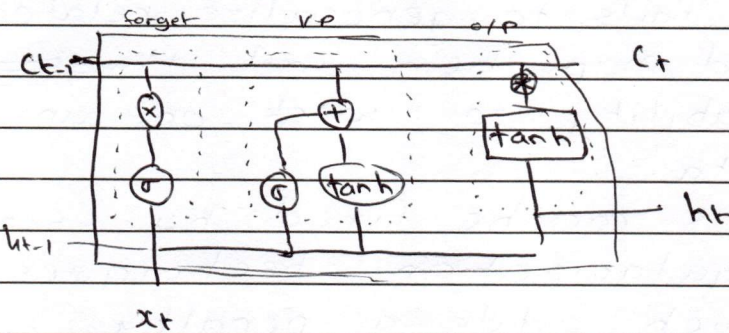
In a depth-wise separable convolution
it will be

$$\text{No. of mul.} = F \times F \times (H-1) \times (W-1) \times (C-1)$$

Depthwise separable convolutions are
widely used in modern architecture
as they rely on multiple channels.

These can be further improved for
parameter efficiency by using
pooling layers like max pooling
or average pooling which helps
reducing the dimension without having
much impact on feature data.



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	Q.3a	① LSTM has a separate channel which stores the cell state (C_t). ② General RNNs only transfer the hidden state, but for longer network, this value often becomes too small due to which the initial data is lost. ③ This leads to the problem of vanishing gradient.  ④ It passes the long memory using the C_t channel along with the hidden state which helps in keeping memory for longer networks. $\frac{\partial L}{\partial h_{t-k}} = \frac{\partial L}{\partial w} \times \frac{\partial w}{\partial h_{t-k}}$



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Question No.	Answer
1	1. The first step in the process of creating a new product is to identify a market need. This is often done through market research, which can involve surveys, focus groups, and other methods of gathering information about potential customers and their needs.
2	2. Once a market need has been identified, the next step is to develop a concept for the new product. This involves brainstorming ideas and creating a rough sketch or prototype of the product.
3	3. The third step is to conduct a feasibility study. This is a detailed analysis of the product concept, including an assessment of the market, the technology, and the financial viability of the project.
4	4. The fourth step is to develop a business plan. This document outlines the company's goals, strategies, and financial projections, and is used to secure funding from investors or lenders.
5	5. The fifth step is to create a prototype. This is a physical model of the product that can be used to test the design and make any necessary adjustments.
6	6. The sixth step is to conduct a pilot test. This involves producing a small batch of the product and testing it in a limited market to gather feedback and assess demand.
7	7. The seventh step is to launch the product. This involves creating a marketing campaign to promote the product and make it available to the general public.
8	8. The eighth step is to monitor the product's performance. This involves tracking sales, customer feedback, and other metrics to ensure the product is meeting its goals and making any necessary adjustments.
9	9. The ninth step is to plan for the future. This involves identifying opportunities for growth and expansion, and developing strategies to capitalize on them.
10	10. The tenth step is to evaluate the overall success of the product. This involves comparing the product's performance against its goals and objectives, and determining whether it is a successful addition to the company's portfolio.

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Q.1b ① Overfitting occurs when the model becomes too complex to learn and it learns all the noisy data as well.

② It show high training accuracy but fails during the testing phase.

③ It typically just memorizes the data instead of learning the features in it.

④ It fails to generalize related feature and map them and thus ~~ing~~ its inability to work on unseen data.

⑤ This can be fixed by using regularization techniques like $L1/L2$ which adds a penalty so that it does not allow high weight for any single feature helping to generalize the input more better.

[illegible]

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Question No.	Answer
1	1. The first step in the process of creating a business plan is to conduct a market research. This involves identifying the target market, understanding the needs and preferences of the customers, and analyzing the competitive landscape. Market research helps in making informed decisions about the products and services to offer, the pricing strategy, and the distribution channels.
2	2. The second step is to develop a business model. This involves determining how the business will generate revenue and sustain itself. It includes identifying the value proposition, the revenue streams, and the cost structure. A clear business model is essential for attracting investors and securing financing.
3	3. The third step is to create a financial plan. This involves projecting the financial performance of the business over a period of time. It includes estimating the startup costs, the operating expenses, the revenue, and the profit. A financial plan provides a clear picture of the business's financial health and helps in making decisions about the funding requirements and the financial goals.
4	4. The fourth step is to develop a marketing and sales strategy. This involves identifying the marketing channels, the sales tactics, and the promotional activities. A well-defined marketing and sales strategy is crucial for reaching the target market, generating leads, and converting them into customers.
5	5. The fifth and final step is to implement the business plan. This involves putting the plan into action, monitoring the progress, and making adjustments as needed. It requires a strong commitment, effective communication, and a flexible approach to adapt to changing market conditions.

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